I. Most Promising Multiple-Access Techniques For 6g

- **NOMA — Non-Orthogonal Multiple Access**
- **RSMA — Rate-Splitting Multiple Access**
- **Cache-Aided MU-MIMO — Cache-Aided Multi-User Multiple-Input Multiple-Output**

Technology	Principle	Potential advantage
OFDMA	Separates users using orthogonal frequency resources	Mature, efficient baseline
NOMA	Allows users to share resources, often distinguished by power	Improved spectrum utilization in favorable conditions
RSMA	Splits messages into common and private streams	Better interference management
SDMA / Massive MIMO	Separates users spatially using antenna arrays	Supports simultaneous users
OTFS-based access	Uses delay-Doppler signal representation	Potential robustness in high-mobility channels
AI-assisted MAC	Learns and adapts resource allocation	Better scheduling and adaptability

** RSMA and NOMA have promising research results, but neither is guaranteed to become the principal 6G access method (ETSI's January 2026 Report).

II. Possible Research Directions Rather Than Established Technologies for 7G

- **AI-native multiple access**: Intelligent systems jointly optimize time, frequency, power, and spatial resources.
- **Semantic-aware multiple access**: Resource allocation considers the importance of transmitted information, not just the number of bits.
- **Integrated communication and sensing access**: Wireless resources are coordinated between data transmission and environmental sensing.
- **Highly distributed spatial access**: Coordinated antenna systems serve users across large distributed networks.